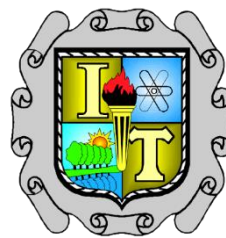




**TECNOLÓGICO
NACIONAL DE MÉXICO**



Tecnológico Nacional de México

Instituto Tecnológico de Saltillo

Ingeniería en Sistemas Computacionales

Cómputo en la Nube

Unidad 4

Práctica 4.3 – Hardening Apache2

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Paso 1: Instalación del Servidor Apache

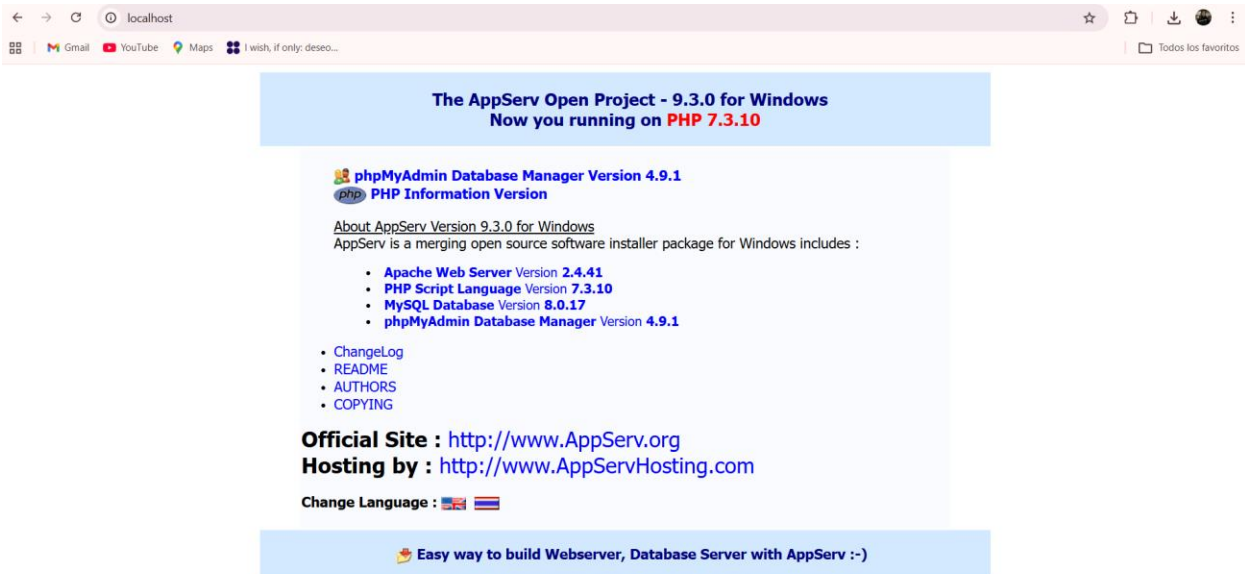
```
silvia@debian-s:~$ sudo apt update
[sudo] contraseña para silvia:
Des:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Des:2 http://security.debian.org/debian-security trixie-security InRelease [43.4 kB]
Des:3 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Des:4 http://deb.debian.org/debian trixie/main Sources [10.5 MB]
```

```
silvia@debian-s:~$ sudo apt install apache2 -y
Upgrading:
  apache2  apache2-bin  apache2-data  apache2-utils

Summary:
  Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 52
  Download size: 2.011 kB
```

```
silvia@debian-s:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-03-20 07:08:29 CST; 1min 10s ago
  Invocation: 2f4288b4c37846629ff876f1df1539bc
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1502 (apache2)
      Tasks: 6 (limit: 2301)
    Memory: 12.8M (peak: 13M)
       CPU: 224ms
    CGroup: /system.slice/apache2.service
```

Paso 2: Verificar funcionamiento en el navegador



Paso 3: Ocultar información sensible

```
GNU nano 8.4 /etc/apache2/conf-available/security.conf *
#ServerTokens Minimal
ServerTokens Prod
#ServerTokens Full

#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | EMail
#ServerSignature Off
ServerSignature Off

#
# Allow TRACE method
#
# Set to "extended" to also reflect the request body (only for testing and
# diagnostic purposes).
#
# Set to one of: On | Off | extended
TraceEnable Off
#TraceEnable On
```

Paso 4: Eliminar módulos innecesarios

```
autoindex
silvia@debian-s:~$ sudo a2dismod autoindex status proxy proxy_http
WARNING: The following essential module will be disabled.
This might result in unexpected behavior and should NOT be done
unless you know exactly what you are doing!
autoindex

To continue type in the phrase 'Yes, do as I say!' or retry by passing '-f': Yes, do as I say!
Module autoindex disabled.
Module status disabled.
Module proxy already disabled
Module proxy_http already disabled
To activate the new configuration, you need to run:
systemctl restart apache2
silvia@debian-s:~$ sudo systemctl restart apache2
silvia@debian-s:~$
```

Paso 5: Configurar HTTPS (SSL) con certificado autofirmado

5.1 Instalar módulos SSL y herramientas

```
silvia@debian-s:~$ sudo apt install openssl ssl-cert -y
ssl-cert ya está en su versión más reciente (1.1.3).
fijado ssl-cert como instalado manualmente.
Upgrading:
libssl3t64 openssl openssl-provider-legacy

Summary:
Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 49
Download size: 4 252 kB
Space needed: 15.4 kB / 16.0 GB available
```

```
silvia@debian-s:~$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
systemctl restart apache2
silvia@debian-s:~$
```

5.2 Crear certificado SSL

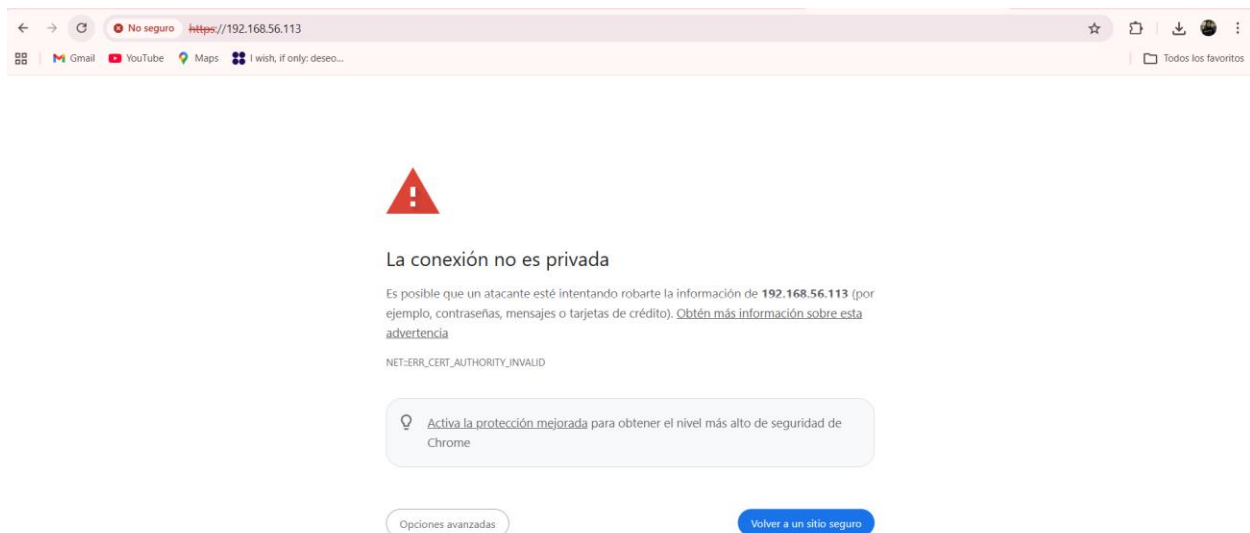
```
-----BEGIN CERTIFICATE-----
MIIDazCCA0GawIBAgIUeYvEH0KQwbCBLJNUcxUmn186+UIwDQYJKoZIhvcNAQEL
BQAwR2lMAkGA1UEBhMCMQVUxwEzARBgNVBAGMCLNvbWUtU3RhdGUxITAFBgNVBAOM
GEludGVybmV0IFdpZ2dpdHMoUHR5IEF0ZDAeFw0yNjAzMjAxMzI1NDRAfW0yNzAz
MjAxMzI1NDRAeUxwEzAJBGNVBAYTAkFVMRMEwEQYDVQIDApTb211LVN0YXRlMSEw
HwYDVQKDBHJbnRlcm5ldCBXaWRnaXRzIFB0eSBMdGQwggeiMA0GCSqGSIb3DQEB
AQUAA4IBDwAwggEKAoIBAQC6G0ynfj2QGCMU8XsmNtVeg/480aKWXWw2bjTq1p1ct
PgZKEJh+qKc0JiZwD65ACK0ynfj2QGCMU8XsmNtVeg/480aKWXWw2bjTq1p1ct
pSZH5Xw4903tB1JwqhPVQ3j/fLXR46DTbyDE0W6GLy3p3LYh/LxL3W9+XiwTb1Gxf
oVKSkvJIQz+1xGcYte3oJFts8KHwBazGZfN8oDS158t5ygsC24VVCVIMDPpErNHPVs
QgVspPrgCZCLlbSt/70E3a/cX0frrtaDyCwsVE9oVu8U60J3yAFHGtt+wwYGotMvu
3CfcJaXJndggzS0i45GeBw6q6TLdgXVzz5EKBYKnocvAgMBAAGjUzBRMB0GA1Ud
DgQWBBOQXAERkx8+xWAOt6j17tpXe+i+8LjAFBgNVHSMEGDAwBgQXAERkx8+xWAOt
6j17tpXe+i+8LjAPBgNVHRMBAF8EBTADAQH/MA0GCSqGSIb3DQEBcwwUAA4IBAQA/
V2zgGI0DZLuI/RGYPp5uWkEC9G+C6RpzKAio+daRQdA0MNgQ2UcBbQ/VEQ50fS
tQf9XCnfjpy9mq6SnnNaXEdYbNODdL6rSUoQL1EZ7nEgw8kPk+nd+geYbuq1Fk1KU
jaTVVuc+MRMMRR7vYK5QVBFl0nrF4g7SRsz3K3DeFwxB97Byvk+wRlwmzX0A2LIP
nVt1EqQzByFAH0zaJZTsmQL9I307C0MappzF7LsMCY+1sNEe0YLCY7/Rdg1awjW
p4eTRC4NB08btCvTzIPsUHqYhGT2sRvx8CsK6/YEjw+eLhtzqfHDe7naBLVWnwdM
2ai6UocpJe1TPniMznmI
-----END CERTIFICATE-----
```

```
GNU nano 8.4 /etc/apache2/sites-available/misitio-ssl.conf
<VirtualHost *:443>
    ServerAdmin webmaster@localhost
    ServerName 192.168.56.113
    DocumentRoot /var/www/html

    SSLEngine on

    SSLCertificateFile /etc/apache2/ssl/apache.crt
    SSLCertificateKeyFile /etc/apache2/ssl/apache.key

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```



6. Firewalls y puertos

```
silvia@debian-s:~$ sudo ss -tuln
Netid      State      Recv-Q     Send-Q               Local Address:Port               Peer Address:Port
udp        UNCONN     0           0                   192.168.56.113:68                 0.0.0.0:*
udp        UNCONN     0           0                   10.0.2.15:68                     0.0.0.0:*
udp        UNCONN     0           0                   [fd00::a00:27ff:fed3:22f]:546     [::]:*
udp        UNCONN     0           0                   [fd00::af66:5a2a:8622:a706]:546    [::]:*
udp        UNCONN     0           0                   [fe80::5f05:ad68:a3e3:2f54]%enp0s8:546 [::]:*
udp        UNCONN     0           0                   [fe80::f43b:3329:799e:32a6]%enp0s3:546 [::]:*
tcp        LISTEN     0           4096                127.0.0.1:41369                   0.0.0.0:*
tcp        LISTEN     0           128                 0.0.0.0:22                       0.0.0.0:*
tcp        LISTEN     0           80                  127.0.0.1:3306                    0.0.0.0:*
tcp        LISTEN     0           511                 *:443                             *:
tcp        LISTEN     0           511                 *:80                              *:
tcp        LISTEN     0           128                 [::]:22                           [::]:*

silvia@debian-s:~$ sudo ufw allow 80
sudo ufw allow 443
sudo ufw enable
Rule added
Rule added (v6)
Rule added
Rule added (v6)
Firewall is active and enabled on system startup
silvia@debian-s:~$
```

7. Revisar logs y monitoreo

```
silvia@debian-s:~$ sudo tail -f /var/log/apache2/access.log

192.168.56.1 - - [20/Mar/2026:07:53:56 -0600] "GET / HTTP/1.1" 200 6069 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:07:53:56 -0600] "GET /icons/openlogo-75.png HTTP/1.1" 200 6046 "https://192.168.56.113/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:07:53:56 -0600] "GET /favicon.ico HTTP/1.1" 404 458 "https://192.168.56.113/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
"
```

```
silvia@debian-s:~$ sudo tail -f /var/log/apache2/error.log
[Fri Mar 20 07:41:57.831252 2026] [mpm_prefork:notice] [pid 2525:tid 2525] AH00170: caught SIGWINCH, shutting
down gracefully
[Fri Mar 20 07:41:58.169983 2026] [mpm_prefork:notice] [pid 2590:tid 2590] AH00163: Apache/2.4.66 (Debian) Ope
nSSL/3.5.5 configured -- resuming normal operations
[Fri Mar 20 07:41:58.170098 2026] [core:notice] [pid 2590:tid 2590] AH00094: Command line: '/usr/sbin/apache2'
[Fri Mar 20 07:53:06.098470 2026] [mpm_prefork:notice] [pid 2590:tid 2590] AH00170: caught SIGWINCH, shutting
down gracefully
[Fri Mar 20 07:53:06.270568 2026] [ssl:warn] [pid 2879:tid 2879] AH01906: 192.168.56.113:443:0 server certific
ate is a CA certificate (BasicConstraints: CA == TRUE !?)
[Fri Mar 20 07:53:06.270676 2026] [ssl:warn] [pid 2879:tid 2879] AH01909: 192.168.56.113:443:0 server certific
ate does NOT include an ID which matches the server name
[Fri Mar 20 07:53:06.351540 2026] [ssl:warn] [pid 2880:tid 2880] AH01906: 192.168.56.113:443:0 server certific
ate is a CA certificate (BasicConstraints: CA == TRUE !?)
[Fri Mar 20 07:53:06.351590 2026] [ssl:warn] [pid 2880:tid 2880] AH01909: 192.168.56.113:443:0 server certific
ate does NOT include an ID which matches the server name
[Fri Mar 20 07:53:06.375172 2026] [mpm_prefork:notice] [pid 2880:tid 2880] AH00163: Apache/2.4.66 (Debian) Ope
nSSL/3.5.5 configured -- resuming normal operations
[Fri Mar 20 07:53:06.375226 2026] [core:notice] [pid 2880:tid 2880] AH00094: Command line: '/usr/sbin/apache2'
```

Conclusiones

Durante la realización de esta práctica se logró configurar correctamente un servidor web Apache en Debian, aplicando medidas de hardening orientadas a mejorar la seguridad del sistema. De igual forma, se implementó el uso del protocolo HTTPS mediante un certificado autofirmado. El navegador mostró una advertencia de seguridad, comprobando que la conexión cifrada funcionó correctamente.

Asimismo, se configuraron puertos y se hizo uso de firewall para permitir únicamente el tráfico necesario, así como la importancia del monitoreo mediante logs para detectar posibles incidencias o comportamientos anómalos.

Esta práctica permitió comprender de manera práctica cómo asegurar un servidor web desde su configuración básica hasta la implementación de medidas de protección más avanzadas, destacando la relevancia del hardening como una estrategia fundamental en la administración de sistemas y en la protección de servicios en la nube.